

Finding the Structure of Words

- Human language is a complicated thing.
- We use it to express our thoughts, and through language, we receive information and infer (understand) its meaning.
- Linguistic expressions may appear to be unorganized, though they actually have an underlying organization or structure.
- Trying to understand a language all together is not a viable approach.
- Linguists have developed a whole discipline that looks at language from different perspectives and at different levels of detail.
- For Example :
- The point of **morphology**, for instance, is to study the variable forms and functions of words.

Finding the Structure of Words

- Its syntax is concerned with the arrangement of words into phrases, clauses, and sentences.
- The rules and limitations governing how words are structured and formed based on their pronunciation are explained and defined by the field of linguistics known as **phonology**.
- The conventions for writing constitute the **orthography** of a language.
- The terms like **etymology** and **lexicology** cover especially the evolution of words and explain the semantic, morphological, and other links among them.
- Words are perhaps the most intuitive units of language, yet they are in general tricky to define.

Finding the Structure of Words

- Knowing how to work with them allows, in particular, the development of **syntactic and semantic abstractions**.
- Hence, finding the structure of words involves two steps:
- First we need to explore **how to identify words** of distinct types in human languages, and
- Second, **how the internal structure of words** can be modelled in connection with the grammatical properties and lexical concepts.
- The discovery of word structure is called **morphological parsing**.

Words and Their Components

- Words are defined in most languages as the smallest linguistic units that can form a complete utterance by themselves.
- The minimal parts of words that deliver aspects of meaning to them are called **morphemes**.
- Words in English are delimited only by whitespace and punctuation (fullstop, comma, and brackets)
- For Example - **Will you read the newspaper? Will you read it? I won't read it.**
- If we make an assumption with insights from etymology and syntax, we notice two words are here: ***newspaper*** and ***won't***.
- The word newspaper has an interesting derivational structure.

Words and Their Components

- In writing, **newspaper** and the associated concept is distinguished from the isolated **news** and **paper**.
- Generally, linguists prefer to analyze **won't** as two syntactic words, or tokens, each of which has its independent role and can be reverted back to its normalized form.
- The structure of **won't** could be parsed as **will** followed by **not**.
- In English, this kind of tokenization and normalization may apply to just a limited set of cases.
- But in other languages, these phenomena have to be treated in a less trivial manner.

Words and Their Components

- In Arabic or Hebrew certain tokens are concatenated in writing with the preceding or the following ones and possibly may change their forms.
- The lexical or syntactic units blur into one compact string of letters and no longer appear as distinct words.
- Tokens behaving in this way can be found in various languages and are often called **clitics**.
- For Example - writing systems of Chinese, Japanese and Thai, whitespace is not used to separate words.
- Tokenization, also known as word segmentation, is the fundamental step of morphological analysis and a prerequisite for most language processing applications.

Words and Their Components

1. Lexemes

- By the term word, we often denote word with not just the one linguistic form in the given context but also as the concept behind the form and the set of alternative forms that can express it.
- Such sets are called lexemes or lexical items and they constitute the lexicon of a language.
- Lexemes can be divided by their behavior into the lexical categories of verbs, nouns, adjectives, conjunctions, articles, or other parts of speech.
- The citation form of a lexeme, by which it is commonly identified, is also called its lemma.

Words and Their Components

- When we transform a lexeme into another one that is morphologically related we say we derive the lexeme:
- for instance, the nouns *receiver* and *reception* are derived from the verb *to receive*.
- For Example - Did you see him? I didn't see him. I didn't see anyone.
- The example presents the problem of tokenization of *didn't* and the investigation of the internal structure of *anyone*.
- The paraphrase of *Did you see him ?* is *I saw no one*
- The lexeme *to see* would be inflected into the form *saw* to reflect its grammatical function of expressing positive past tense.
- Likewise, *him* is the oblique case form of *he*.

Words and Their Components

- In the paraphrase, *no one* can be perceived as the minimal word synonymous with *nobody*.

2. Morphemes

- Morphology is the study of the structure of words i.e the way words are built up with smaller and minimal units of meaning which are termed as **morphemes**.
- For Example
- **Played** = play ed
- **Cats** = cat s
- The word played has two morphemes: play ‘word’ and ed ‘plural marker’.

Words and Their Components

- The word cat has two morphemes: cat ‘word’ and s ‘plural marker’.
- **There are two broad types of morphemes :**
 - i. Stem (Root) ii. Affixes
- **Root** is the main meaning bearing morpheme of the word.
- Example : play, cat, friend etc.
- **Affixes** add ‘additional’ meanings of different kinds.
- Example : -ed, -s, un-, -ly etc.
- Two main types of affixes:
 - i. Prefixes precede the stem: un-, in- etc.
 - ii. Suffixes follow the stem: -ed, -s, un-, -ly, etc.

Words and Their Components

- **Affixes are called bound morphemes** as they cannot occur on their own and must combine with a root/stem.
- There are two basic processes of word formation :

i. Inflection

ii. Derivation

- Inflection is a process where affixes are added to a root/stem to perform some grammatical functions but the category of the word remains the same.
- Example

Lemma	Singular	Plural
cat	cat	cats
knife	knife	knives

- The process through which the new words are formed by adding an affix to an existing word is called derivation. Example – inter+national = international.

Words and Their Components

- Unlike inflection, derivation often leads to change in the category.
- The simplest morphological process concatenates morphs one by one.
- For Example —
- The word **dis-agree-ment-s**, where **agree** is a free lexical morpheme and the other elements are bound grammatical morphemes contributing some partial meaning to the whole word.
- The alternative forms of a morpheme are termed allomorphs.

Words and Their Components

3. Typology

- Morphological typology is a linguistic classification system that categorizes languages based on the ways they use morphemes, which are the smallest units of meaning in language.
- It can consider various criteria, and during the history of linguistics, different classifications have been proposed.
- Based on quantitative relations between words, their morphemes, and their features, the primary morphological typological categories are :

A. Isolating or Analytical languages

- In isolating languages, words are typically composed of one or more free morphemes, and there is little or no use of bound morphemes like prefixes or suffixes to indicate grammatical relationships.

Words and Their Components

- Each word often carries a single, specific meaning.
- Example - typical isolating members are Chinese, Vietnamese, and Thai.

B. Synthetic languages

- Synthetic languages can combine more morphemes in one word and are further divided into agglutinative and fusional languages.

i. Agglutinative languages

- Agglutinative languages use a high number of bound morphemes, each of which typically carries a single grammatical meaning.
- Example – Japanese language is an agglutinative language where suffixes are added to roots to convey aspects of tense, mood, and case.

Words and Their Components

ii. Fusional Languages

- Fusional languages use bound morphemes, but these morphemes often carry multiple grammatical meanings or features simultaneously.
- The morphemes are fused together, making it more challenging to separate individual meanings.
- Example - Arabic, Latin, Sanskrit, German use fusional languages.
- In addition with the word formation processes mentioned above, we can also find out languages using concatenative and nonlinear forms.
- **Concatenative languages** link morphs and morphemes one after another and **Nonlinear languages** allowing structural components to merge non-sequentially to apply tonal morphemes or change the consonantal of words.

Issues and Challenges

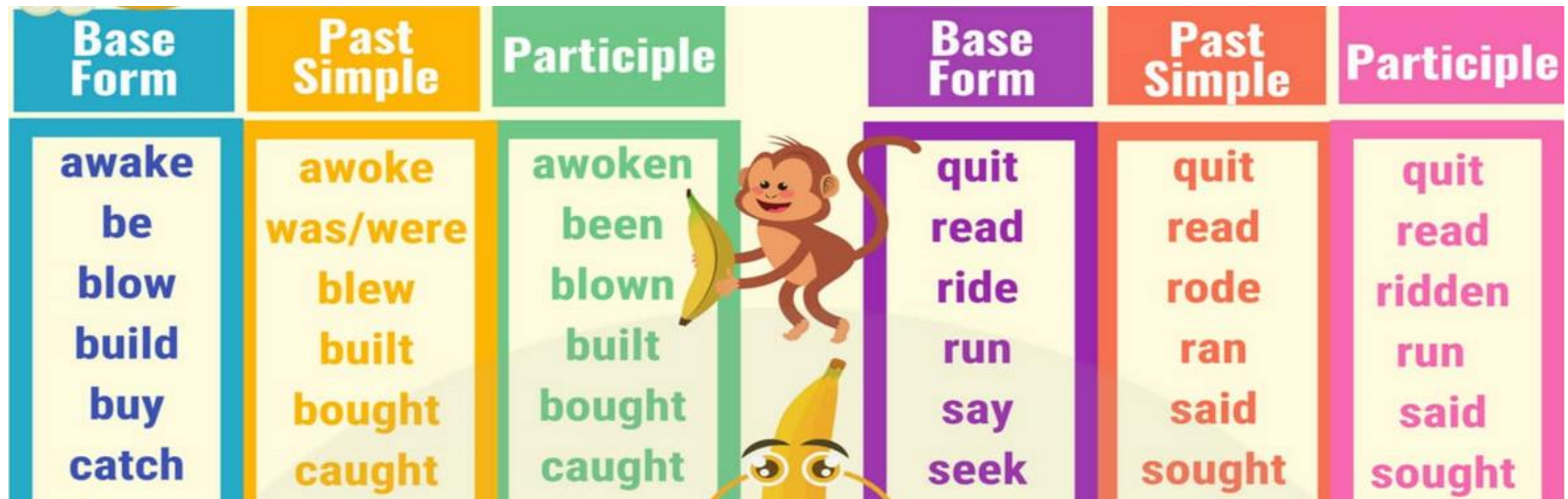
- Issues and challenges related to words and their components, which encompass morphemes, phonemes, and other linguistic elements which are important considerations in the field of linguistics and natural language processing.
- Here are some key issues and challenges:
 1. **Irregularity** word forms are not described by a prototypical linguistic model.
 2. **Ambiguity** word forms be understood in multiple ways out of the context of their discourse.
 3. **Productivity** is the inventory of words in a language finite, or is it unlimited ?

Issues and Challenges

- **Irregularity**
- The phenomenon where certain words or word forms does not follow regular patterns or rules interms of their morphology or syntax.
- By irregularity, we mean existence of such forms and structures that are not described appropriately by a prototypical linguistic model.
- Some irregularities can be redesigned and improve its rules, but other lexically dependent irregularities often cannot be generalized.
- It is a challenge for the algorithms which follow particular patterns.
- What are actually the word forms that form irregularity ?
- There are many word forms that form irregularity like:

Issues and Challenges

- i. Irregular verb or nouns – which does not follow standard pattern of inflection.



Base Form	Past Simple	Participle	Base Form	Past Simple	Participle
awake	awoke	awoken	quit	quit	quit
be	was/were	been	read	read	read
blow	blew	blown	ride	rode	ridden
build	built	built	run	ran	run
buy	bought	bought	say	said	said
catch	caught	caught	seek	sought	sought

- ii. Exceptional Inflection – is mainly caused by comparative and superlative adjectives. For example

	Comparative	Superlative
Big	Bigger	Biggest
Dark	Darker	Darkest
Good	Better	Best

Issues and Challenges

2. Ambiguity

- Words forms that can be understood in multiple ways out of the context.
- Word forms that look same but have distinct functions or meaning also called as homonyms.
- Ambiguity arises in morphological processing and language processing.
- Four kinds of ambiguity are :
 - i. **Word sense ambiguity** – A particular word will be having different meanings depending on the context in which they are used.
- For Example

Bank has ambiguity relating to money or river bank.

Bat has ambiguity relating to a bird or cricket bat.

Issues and Challenges

ii. Parts of speech ambiguity – The parts of speech of a particular word will be changing in different context.

For Example - I run (verb)

He went for a run (noun)

iii. Structural ambiguity – Structural ambiguity is having with multiple valid syntactic structures.

A sentence which can't be written in one particular form. The sentence is written in multiple ways which give same meaning.

For example - “She walked gracefully through the garden.”

Sentence 1 - Through the garden, she walked with grace.

Sentence 2 - She strolled elegantly in the garden.

Issues and Challenges

iv. Referential ambiguity – In referential ambiguity, the name of a person or thing is reference by pronouns like he/she/it/they/this etc.

For Example - John went to the store, and he bought some groceries.

I found a book at the library, and it was really interesting.

3. Productivity

- Productivity refers to the ability to generate new words or word forms using productive rules.
- For Example – According to Wikipedia, **googol** means 1 followed by 100 zeroes.
- From googol, an unknown word google was generated and by using some productivity rules, a new words are generated like googling, googlish, googleology etc.

Morphological Models

- There are many possible approaches to designing and implementing morphological models.
- Over time, computational linguistics has witnessed the development of a number of formalisms and frameworks.
- The most prominent types of computational approaches to morphology are:
 1. Dictionary Lookup
 2. Finite-State Morphology
 3. Unification-Based Morphology
 4. Functional Morphology

Morphological Models

1. Dictionary Lookup

- Dictionary Lookup, also known as Lexicon-Based Morphological Analysis that relies on pre-built dictionaries or lexicons to analyze and process words.
- A dictionary is understood as a data structure that directly enables obtaining some precomputed results, in our case word analyses.
- The data structure can be optimized for efficient lookup, and the results can be shared.
- Lookup operations are relatively simple and are usually quick.

Morphological Models

- Dictionaries can be implemented, for instance, as lists, binary search trees, hash tables, and so on.
- Dictionary Lookup as a morphological model works with following ways:
 - i. **Dictionary Creation:** In this approach, a comprehensive dictionary or lexicon is compiled for the target language.
 - ii. **Word Analysis:** When a word is encountered in text, the Dictionary Lookup model first attempts to find an exact match for the word in the dictionary.
 - iii. **Lemma Retrieval:** The model then retrieves the lemma (base form) of the word from the dictionary..

Morphological Models

iv. Handling Ambiguity: Dictionary Lookup models may need to handle cases of word ambiguity, where a single word form can have multiple possible lemmas or meanings.

- **Advantages of Dictionary Lookup are :**

- a. Accuracy

- b. Transparency

- **Limitations of Dictionary Lookup are :**

- a. Limited Coverage

- b. Ambiguity Handling

- c. Resource-Intensive

Morphological Models

2. Finite-State Morphology

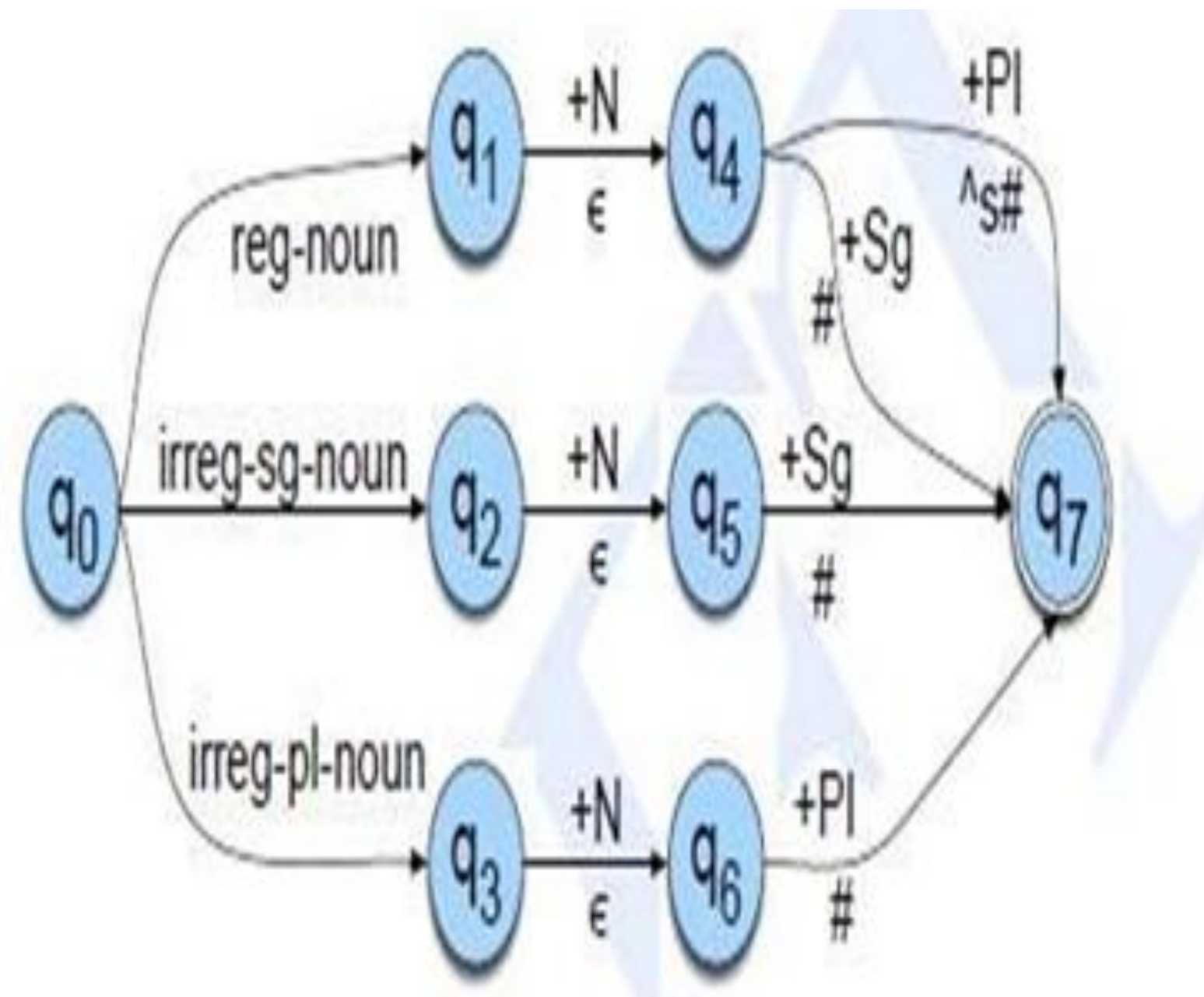
- By finite-state morphological models, we mean those in which the specifications written by human programmers are directly compiled into finite-state transducers.
- The two most popular tools supporting this approach, XFST (Xerox Finite-State Tool) and Lex Tools.
- Finite-state transducers are computational devices extending the power of finite-state automata.
- FST consist of a finite set of nodes connected by directed edges labeled with pairs of input and output symbols.
- In such a network or graph, nodes are also called states, while edges are called arcs.

Morphological Models

- Traversing the network from the set of initial states to the set of final states along the arcs is equivalent to reading the sequences of encountered input symbols and writing the sequences of corresponding output symbols.
- The set of possible sequences accepted by the transducer defines the input language;
- The set of possible sequences emitted by the transducer defines the output language.
- Following example shows the FST state diagram for the input words and their corresponding morphological parsed output or morphological parsing.

Morphological Models

Input	Input Morphological parsed output
Cats	cat +N +PL
Cat	cat +N +SG
Cities	city +N +PL
Geese	goose +N +PL
Goose	goose +N +SG) or (goose +V)
Geese	goose +V +3SG
Merging	merge +V +PRES-PART
Caught	(caught +V +PAST-PART) or (catch +V +PAST)



- Figure : A schematic finite state transducer for English number inflection T_{noun} . The symbols above each arc represent elements of the morphological parse in the lexical tape.

Morphological Models

- In finite-state computational morphology, it is common to refer to the input word forms as **surface strings** and to the output descriptions as **lexical strings**.
- In English, a finite-state transducer could analyze the surface string children into the lexical string child [+plural].
- for instance - woman from woman [+plural(women)]
- Relations on languages can also be viewed as functions.
- Let us have a relation R , and let us denote by $[\Sigma]$ the set of all sequences over some set of symbols Σ .
- The domain and the range of R are subsets of $[\Sigma]$.

Morphological Models

- We can then consider R as a function mapping an input string into a set of output strings, formally denoted by this type signature, where $[\Sigma]$ equals *String*.

$$R :: [\Sigma] \rightarrow \{[\Sigma]\}$$

$$R :: \textit{String} \rightarrow \{\textit{String}\}$$

- A theoretical limitation of finite-state models of morphology is the problem of capturing **reduplication** of words or their elements (e.g., to express plurality) found in several human languages.

Morphological Models

3. Unification-based morphology

- Unification-based morphology is a computational approach to morphological analysis and generation that uses unification to **combine information about the morphemes and features of a word.**
- Unification is a logical operation that merges two or more sets of constraints into a single consistent set.
- This approach focuses on the relationships between morphemes (the smallest units of meaning in a language) and how they combine to create word forms.

Morphological Models

- The key components and concepts of Unification-Based Morphology are :

1. Feature Structures:

- Unification-Based Morphology represents linguistic information using feature structures, which are data structures that consist of attribute-value pairs.
- These feature structures encode information about morphemes, their grammatical properties, and their relationships within words.

2. Morphemes and Morphological Rules : Morphemes are the building blocks of words, representing units of meaning.

Morphological Models

- Unification-Based Morphology defines morphological rules that specify how morphemes can combine and interact.
- These rules are expressed using feature structures.
- **For Example :**
- In English, the word "cats" consists of two morphemes: "cat" and "-s" (indicating plural).
- These morphemes can be represented as feature structures:
- Morpheme "cat": {base: "cat", pos: "noun", number: "singular"}
- Morpheme "-s": {base: "", pos: "plural marker", number: "plural"}

Morphological Models

3. Lexical Entries:

Each word in a language is associated with a lexical entry, which includes information about the word's morphemes, their features, and how they are combined.

Lexical entries are represented as feature structures.

For Example :

Lexical Entry for "cats":

Word: "cats"

Morphemes: {"cat", "-s"}

Feature Structures:

{ { base: "cat", pos: "noun", number: "singular"},

{ base: "-", pos: "plural marker", number: "plural"} }

Morphological Models

4. Morphological Analysis:

- Unification-Based Morphology performs morphological analysis by unifying feature structures representing morphemes to generate the complete word form.
- This process involves unification, which combines feature structures while preserving shared features and resolving conflicts.

- For Example :

To analyze the word "cats," the feature structures for its morphemes are unified to generate the complete word form:

- Unified Feature Structure for "cats":

`{base: "cat", pos: "noun", number: "plural"}`

Morphological Models

5. Morphological Generation:

- Morphological generation, involves starting with feature structures representing the desired grammatical and semantic properties of a word and generating the word form by applying morphological rules in reverse.
- **For Example:**
- Conversely, in morphological generation, we can start with the desired features and generate a word form:
- Desired Feature Structure: {base: "dog", pos: "noun", number: "plural"}
- Morphological Rules:
 - Apply "-s" to indicate plural.
- Generated Word Form: "dogs"

Morphological Models

6. Ambiguity Handling

- Unification-Based Morphology provides a framework for handling morphological ambiguity by representing multiple potential feature structures and allowing for disambiguation based on context or additional linguistic constraints.
- Consider the word "saw," which could be a past tense verb or a noun (e.g., a tool).
- Ambiguity Representation:
 - Past Tense Verb: {base: "see", pos: "verb", tense: "past"}
 - Noun (Tool): {base: "saw", pos: "noun"}

Morphological Models

4. Functional Morphology

- Functional morphology defines its models using principles of functional programming and type theory.
- It treats morphological operations and processes as pure mathematical functions and organizes the linguistic expression.
- It also abstract elements of a model into distinct types of values and type classes.
- Functional morphology is not limited to modelling particular types of morphologies in human languages, it is also especially useful for fusional morphologies.

Morphological Models

- Linguistic notions like paradigms, rules, exceptions, grammatical categories, parameters, lexemes, morphemes, and morphs can be represented intuitively and clearly.
- Functional morphology implementations are intended to be reused as programming libraries capable of handling the complete morphology of a language.
- We can describe inflection I , derivation D , and lookup L as functions of these generic type as below :
- $I :: \textit{lexeme} \rightarrow \{\textit{parameter}\} \rightarrow \textit{form}$
- $D :: \textit{lexeme} \rightarrow \{\textit{parameter}\} \rightarrow \textit{lexeme}$
- $L :: \textit{content} \rightarrow \{\textit{lexeme}\}$

Morphological Models

- Many functional morphology implementations are embedded in a general-purpose programming language.
- This gives programmers more freedom with advanced programming techniques and allows them to develop full-featured, real-world applications for their models.
- For instance
- The Zen toolkit for Sanskrit morphology is written in OCaml.

Morphological Models

- It influenced the functional morphology framework in Haskell, with which morphologies of Latin, Swedish, Spanish, Urdu, and other languages have been implemented.
- Morphological grammars in Grammatical Framework can be extended with descriptions of the syntax and semantics of a language.
- Grammatical Framework itself supports multi-linguality, and models of more than a dozen languages that are available as open-source software.

Assignment Questions

1. Define the following terminologies
 - a. Morphology
 - b. Phonology
 - c. Orthography
 - d. Etymology and lexicology
 - e. Morphological Parsing
2. Give the steps to find the structure of the words given a sentence.
3. What is the significance of words and their components.
4. Explain the key components of word with example.
5. What are the issues and challenges in finding the structure o words ? Discuss.
6. Explain the dictionary lookup as a morphological model.
7. What is a finite state morphology ? Traverse the given words using FST
 - i. cities
 - ii. Cats
 - iii. Gooses
 - iv. Caught
 - v. runs
 - vi. Feet
8. What is unification-based morphology ? Give its key components with examples.
9. Define functional morphology. What is the significance of using functional morphology.